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Development of rapid clonal propagation method in *Rauwolfia serpentina*(L). Benth. ex Kurz .

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Abstract : In view of the growing world population, increasing anthropogenic activities, rapidly eroding natural ecosystem, the natural habitat for a great number of herbs and trees are dwindling. Many of them are facing extinction. With an ever increasing inclination for the herbal medicine, there is an obligatory demand for the huge raw material of medicinal plants thus contributing to the great extinction risk to many medicinal plants and obviously the loss of genetic diversity. Hence the intervention of biotechnology is need of the hour for accelerating large scale clonal multiplication of the medicinal plants and their conservation through Tissue banks and Gene Banks. It will further help in in vitro selection of lines with desirable morphological traits such as enhanced production of secondary metabolites. *Rauwolfia serpentina* (Apocynaceae) a woody perennial shrub is one of the most important medicinal plant used in India for the treatment of snakebites, hypertension, high blood pressure and mental illness. The present investigation is an effort to establish rapid clonal propagation method in *Rauwolfia serpentina* using nodal and apical buds of mature plants. The explants were cultured on MS media supplemented with varying concentrations of Kinetin, BAP and 2,4-dichlorophenoxyacetic acid. The best multiple shooting response was observed in media supplemented with BAP along with low concentration of auxin. The developed shoots were excised after one month and were cultured on half strength MS media supplemented with different concentration of IBA. Maximum rooting was observed (80%) in 0.5 mg/ml of IBA. Plants with well developed roots were acclimatized successfully in small pots.

Key Words: *Rauwolfia serpentina*, clonal propagation, Kinetin, BAP, 2,4-D, IBA.

INTRODUCTION

Human beings have been utilizing plants for basic preventive and curative health care since time immemorial. In India, many indigenous plants are used in herbal medicine to cure diseases and heal injuries. In view of the growing world population, increasing anthropogenic activities, rapidly eroding natural ecosystem etc., the natural habitat for a great number of herbs and trees are dwindling. This makes human activity the most important source of the current decline in biodiversity.

To cope up with this alarming situation, the recent exciting developments in biotechnology have come as a boon. The powerful techniques in plant cell and tissue culture, recombinant DNA, bioprocess technologies etc have offered mankind the great potency of exploiting the totipotent biosynthetic and biotransformation capabilities of plant cells under *in vitro* conditions.

Rauwolfia serpentina L. (family: Apocynaceae) is a woody perennial shrub, commonly known with different names; sarpagandha, snake root plant, chotachand, chandrika, etc. The roots of this plant have been used for centuries in ayurvedic medicines by different ethnic groups as an antidote to the stings of insects and poisonous reptile and for the treatment of mental disease, epilepsy, sleeplessness, hypertension, blood pressure, gastrointestinal diseases, circulatory disorders, pneumonia, fever, malaria, asthma, skin diseases, scabies, eye diseases, spleen diseases, AIDS, rheumatism, body pain, veterinary diseases etc.(1). It has also been used as febrifuge and stimulant to uterine contraction for insomnia and most of all for insanity. High quantity of flavonoids, saponins and

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alkaloids were found on *Rauwolfia serpentina* (2). Reserpine is the most important alkaloid present in root, stem and leaves of the plant. 90% of the total reserpine content was produced from the root, with the stem and leaf containing 10%, it proves the tissue specificity of the reserpine production (3). In some reports it has been shown that the regenerated shoots exhibited three main alkaloids viz. yohimbine (0.0183%), ajmaline (0.0763%), and ajmalicine (0.0794%). Reserpine was also present in a small quantity (0.00801%) (4).

The plant is vegetatively propogated by root cutting due to production of large number of non-viable seeds and low germination percentage. Further *Rauwolfia* is threatened with extinction due to its limited cultivation and over exploitation by local people, government agencies and various pharmaceutical houses. Due to the prevailing reasons there is a huge need for *in vitro* propagation of *Rauwolfia serpentina* to satisfy the growing commercial demand of the plant for the production of life supporting alkaloids and conservation of this valuable endangered plant itself. There are reports of *in vitro* propagation and manipulation of this plant (5 & 6). Maximum multiple shoots (85.6%) were observed on MS medium supplemented with 5.0 mg/1 BAP and 0.5 mg/1 IBA along with 2.5% sucrose and 0.85% agar and maximum rooting (76.6%) in 0.5 mg/1 IBA (5). It was reported that the combination of IBA (0.125 mg/L.) + BAP (1.0 mg/L.) produced better results for both callus induction and direct regeneration (6). Improvements in plant tissue culture techniques for the mass propagation of *R. serpentina* are highly desirable. Therefore, in the present investigation apical and axillary buds of *Rauwolfia serpentina* were cultured on various combination of hormonal media. Multiple shooting was observed in MS media supplemented with BAP alone.

MATERIALS & METHODS

The disease free, young and healthy apical and nodal explants were selected (as young cells are supposed to have retained their totipotency). Endophytic microflora is a major cause of contamination for *in vitro* culture of *R. serpentina*. Hence, a proper sterilization was needed for successful culture. Axilliary buds were surface sterilized with 0.1 % $HgCl_2$ for 5 min and repeatedly washed with sterile water for 3-4 times before inoculation on MS medium supplemented with various phytohormones under aseptic conditions. Then the inoculated culture tubes were then transferred to Culture Room under the standard conditions of temperature (25 ± 2ºC) for 16 hours of daybreak and 8 hours night break under the cool white fluorescent light. Observations were recorded after every 5 days. Shoot regeneration from buds was observed after 4 weeks. Well developed elongated shoots were excised and cultured on root induction medium (MS ½ strength + 0.5-1 mg/l of IBA).

Table 1 Effect of different hormones on axillary bud culture of *Rauwolfia serpentina*

	Hormones (mg/l)	Percentage of shoot response	No. of shoots per explant	Hormones (mg/l)	Percentage of shoot response	No. of shoots per explant		
1	2,4-D	BAP			2,4-D	Kinetin		
2	0.125	1	25%	1-2	0.125	1	-	-
3	0.125	1.5	56%	2-3	0.125	1.5	24.2%	1
4	0.125	2	70%	3-4	0.125	2	40%	1-2
5	0.250	1	33%	0-1	0.250	1	34%	1-2
6	0.250	1.5	40%	0-1	0.250	1.5	29%	1-2
7	0.250	2	45%	1-2	0.250	2	35%	1
8	--	1	57%	2-3	--	1	45%	1
9	--	1.5	85 %	4-5	--	1.5	42%	1-2
10	--	2	68%	3-4	--	2	53%	1-2

OBSERVATION

Table 1- Shooting

Explant	Phytohormone(mg/l) 2,4-D	Regeneration BAP	Kinetin	No. of shoot /explant	
	0.125	-	-	-	-
	0.250	-	-	-	-
	0.125	1	-	++	2-3
	0.125	2	-	+++	3-4
	0.250	1	-	+	1-2
	0.250	2	-	++	2-3
	-	1	-	+++	3-4
	-	2	-	++++	4-5
	0.125	-	1	++	2-3
	0.125	-	2	+++	3-4
	0.250	-	1	+	1-2
	0.250	-	2	++	2-3
	-	-	1	++	2-3
	-	-	2	+++	3-4



Fig 1(a) Axillary buds growing on MS media supplemented with BAP (1.5 mg/l) showing multiple shooting.

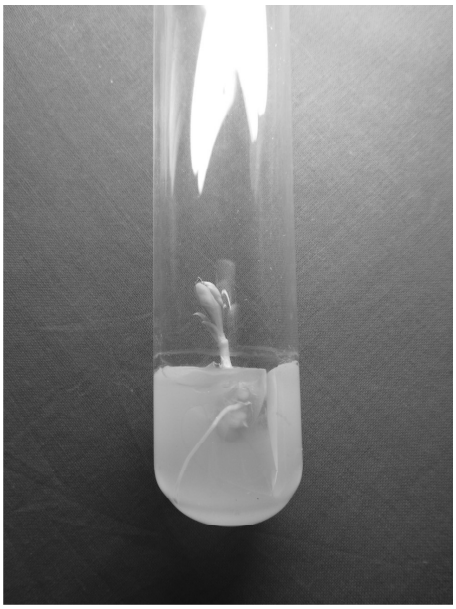


Fig 1 (b) Excised shoots subcultured on fresh MS media supplemented with BAP (1.5 mg/l).

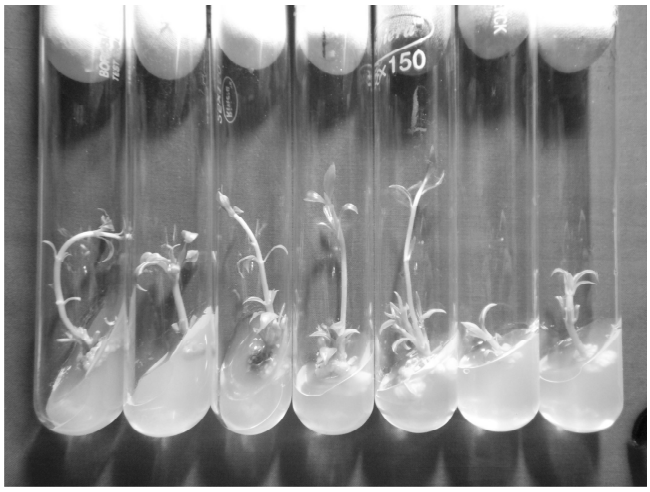


Fig 1(c) Well developed shoots on Root Induction media

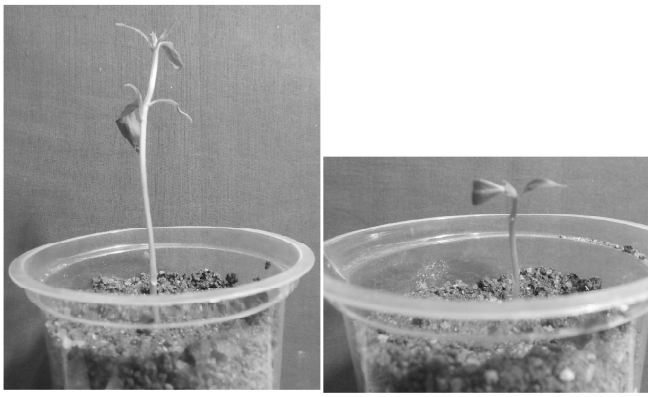


Fig 1(d) *in vitro* raised plants growing successfully in pots.

RESULT AND DISCUSSION

Two different combination of hormones, namely 2,4-D + BAP and 2,4 D + Kinetin in varied concentrations were used. Sprouting was observed in all the horomone combinations including MS alone though in varied percentage. Though MS basal media is effective in bud break, the buds didn't grow further. MS media supplemented with 2,4-D and BAP and kinetin showed better results in terms of sprouting as well as number of buds per explant. Higher percentage of bud break was observed in MS with BAP and MS with BAP and 2,4-D at lower concentrations. But considering the number of shoots per explant BAP alone yielded better result followed by BAP 1.5 mg/L + 2,4-D (0.125mg/l) fig 1 (a). It was observed that 2,4-D alone promoted profuse callusing which affected the of growth and elongation of shoot. However when supplemented with both BAP and Kinetin it shows better regeneration. The auxin:cytokinin ratio regulate morphogenesis in culture tissue. The effects of auxins and cytokinins on shoot multiplication of various medicinal plants have been reported. It has shown that 6-Benzylaminopurine (BAP), at high concentration (1–5ppm), stimulates the development of the axillary meristems and shoot tips of *Atropa belladona* (7). It has been observed that cytokinin is required, in optimal quantity, for shoot proliferation in many genotypes but inclusion of low concentration of auxins along with cytokinin triggers the rate of shoot proliferation (8, 9and 10). However in our set of experiments MS media supplemented with 6-Benzylaminopurine (BAP) alone at concentration of 1.5-2 mg/L showed very good regeneration of 70-85% fig 1(a). MS media supplemented BAP (1.5 mg/l) showed multiple shooting (4-5 shoots/per explants) (table 1, fig. 1a). Although kinetin is reported to promote shoot bud initiation in nodal explants of many plant species, BAP is most efficient cytokinin for axillary bud initiation and proliferation (11). In MS media supplemented with Kinetin only much elongation of the shoot was observed with few no. of shoots as compared to media supplemented with BAP alone. Among all the combinations of phytohormones in both apical and nodal explants the best results of direct regeneration were observed in media supplemented with BAP (1.5mg/l) followed by 2,4-D(0.125) + BAP(2). Manipulation of cytokinin and auxin concentration was found to be very effective in large scale propagation of nodal explants.

For initiation of roots in the explant, MS half strength supplemented with auxins has been reported to be very effective in many medicinal plants (12). The well developed and elongated shoots were excised (fig. 1 b, 1c) and individually transferred to root induction media MS(½ strength) supplemented with IBA of two different concentration (0.5mg/l and 1mg/l). The explants developed roots 10 days after inoculation on root induction medium (fig. 1c)). Among the different concentration of media, the best result of root formation was observed in media

containing MS+IBA (0.5mg/l). The *in vitro* raised plants were then transferred in small pots with autoclaved soil (fig 1d). These plants were acclimatized successfully. The survival rate of plants was 80% and the *in vitro* raised plants didnt show any altered morphology when compared with the field grown plants.

In-vitro propagation of plants holds tremendous potential for the production of high-quality plant-based medicines. This can be achieved through different methods including micropropagation. Micropropagation has many advantages over conventional methods of vegetative propagation, which suffer from several limitations. With micropropagation, the multiplication rate is greatly increased. It also permits the production of pathogen-free material. (13). Micropropagation of various plant species, including many medicinal plants, has been reported. Propagation from existing meristems yields plants that are genetically identical with the donor plants.

In the present investigation a protocol for rapid clonal propagation of *Rauwolfia* spp. through axillary buds have been reported. As observed in the present study the apical and nodal bud may be used for the mass propagation of *Rauwolfia serpentina* through micropropagation technique. The protocol promises a high regeneration and high survival rate for the large scale propagation of *Rauwolfia.* These tissue culture raised plants may be utilized for production of secondary metabolites and it will be a more reliable, predictable and simpler process as compared to extraction from *in vivo* raised plants. Extraction from the *in vitro* tissues is much simpler than extraction from organized, complex tissues of a plant. Plant tissue culture techniques offer the rare opportunity to tailor the chemical profile of a phytochemical product, by manipulation of the chemical or physical microenvironment, to produce a compound of potentially more value for human use.

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